

PRAKTIKUM SISTEM OPERASI

MODUL IX

SYSCALL



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

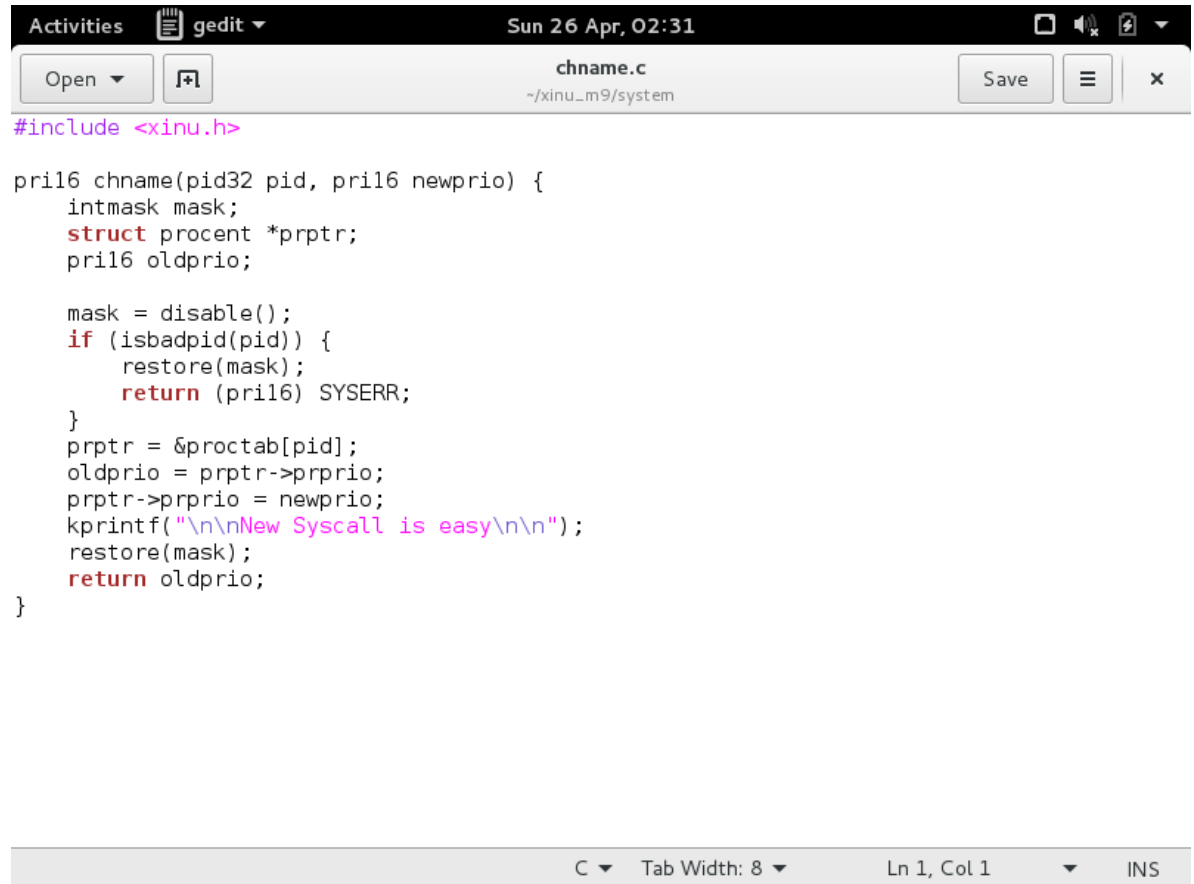
UNIVERSITAS TELKOM

2026

JURNAL

1. Screenshot kode chname.c dan hasil runnya di minicom.

Jawaban:



```
#include <xinu.h>

pri16 chname(pid32 pid, pri16 newprio) {
    intmask mask;
    struct procent *prptr;
    pri16 oldprio;

    mask = disable();
    if (isbadpid(pid)) {
        restore(mask);
        return (pri16) SYSERR;
    }
    prptr = &proctab[pid];
    oldprio = prptr->prprio;
    prptr->prprio = newprio;
    kprintf("\n\nNew Syscall is easy\n\n");
    restore(mask);
    return oldprio;
}
```

C ▾ Tab Width: 8 ▾ Ln 1, Col 1 ▾ INS

```
Activities Terminal Sun 26 Apr, 02:43
xinu@xinu-develop-end: ~/xinu_m9/compile

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process    recv   20   3 0x005E7FFC 0x005E7F44 65536
5 shell           recv   50   4 0x005D7FFC 0x005D79AC 8192
6 ps              curr   20   5 0x005F7FFC 0x005F7FC4 8192

xsh $ uptime
Xinu has been up 1 second(s)

New Syscall is easy

New Syscall is easy

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

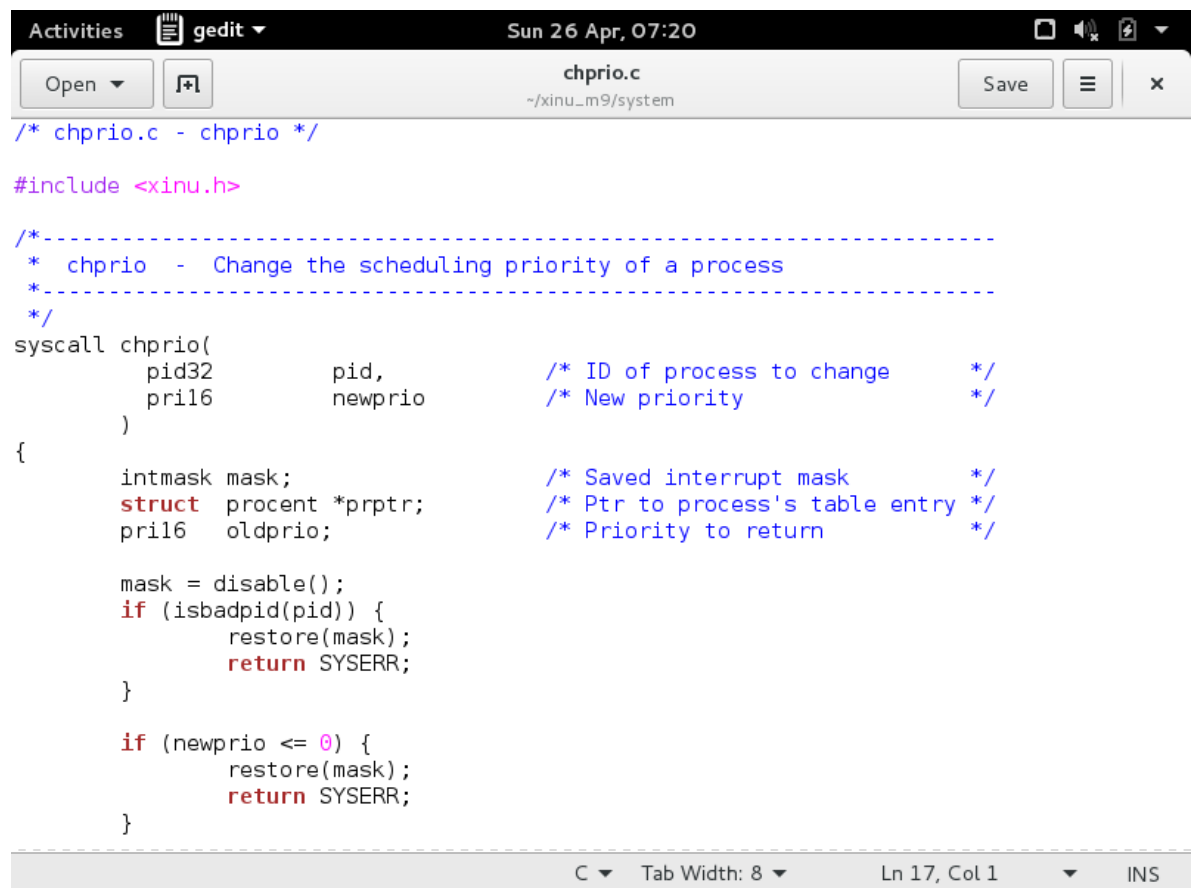
- a. Buat file `chname.c` baru di folder `xinu_m9/system/` Masuk ke folder shell: `cd ~/xinu_m7/shell`
- b. Tulis kode syscall baru:

```
1. #include <xinu.h>
2.
3. pri16 chname(pid32 pid, pri16 newprio) {
4.     intmask mask;
5.     struct procent *prptr;
6.     pri16 oldprio;
7.
8.     mask = disable();
9.     if (isbadpid(pid)) {
10.         restore(mask);
11.         return (pri16) SYSERR;
12.     }
13.     prptr = &proctab[pid];
14.     oldprio = prptr->prprio;
15.     prptr->prprio = newprio;
16.     kprintf("\n\nNew Syscall is easy\n\n");
17.     restore(mask);
18.     return oldprio;
19. }
20.
```

- c. Daftarkan prototype di xinu_m9/include/prototypes.h
- d. Compile dengan make clean lalu make
- e. Jalankan sudo minicom
- f. Jalankan perintah uptime
- g. Muncul output "New Syscall is easy"

2. Perbaiki chprio.c agar validasi id (0 sampai NPROC) dan prioritas (harus positif), lalu compile dan screenshot hasilnya.

Jawaban:



```

/* chprio.c - chprio */

#include <xinu.h>

/*-----
 * chprio - Change the scheduling priority of a process
 *-----
 */
syscall chprio(
    pid32    pid,          /* ID of process to change */
    pri16    newprio       /* New priority */
)
{
    intmask mask;
    struct procent *prptr;
    pri16 oldprio;          /* Saved interrupt mask */
                          /* Ptr to process's table entry */
                          /* Priority to return */

    mask = disable();
    if (isbadpid(pid)) {
        restore(mask);
        return SYSERR;
    }

    if (newprio <= 0) {
        restore(mask);
        return SYSERR;
    }
}

```

C Tab Width: 8 Ln 17, Col 1 INS

```
/*  
syscall chprio(  
    pid32    pid,  
    pri16    newprio  
)  
{  
    intmask mask;  
    struct procent *prptr;  
    pri16 oldprio;  
  
    mask = disable();  
    if (isbadpid(pid)) {  
        restore(mask);  
        return SYSERR;  
    }  
  
    if (newprio <= 0) {  
        restore(mask);  
        return SYSERR;  
    }  
  
    prptr = &proctab[pid];  
    oldprio = prptr->prprio;  
    prptr->prprio = newprio;  
    restore(mask);  
    return 0xffff & oldprio;  
}
```

- a. Buka file chprio.c di folder xinu_m9/system/
- b. Tambahkan validasi newprio <= 0 setelah validasi isbadpid
- c. Kode chprio.c yang sudah diperbaiki:

```
1. /* chprio.c - chprio */  
2. #include <xinu.h>  
3.  
4. syscall chprio(  
5.     pid32    pid,  
6.     pri16    newprio  
7. )  
8. {  
9.     intmask mask;  
10.    struct procent *prptr;  
11.    pri16 oldprio;  
12.  
13.    mask = disable();  
14.    if (isbadpid(pid)) {  
15.        restore(mask);  
16.        return SYSERR;  
17.    }  
18.    if (newprio <= 0) {  
19.        restore(mask);
```

```

20.     return SYSERR;
21. }
22. prptr = &proctab[pid];
23. oldprio = prptr->prprio;
24. prptr->prprio = newprio;
25. restore(mask);
26. return 0xffff & oldprio;
27. }
28.

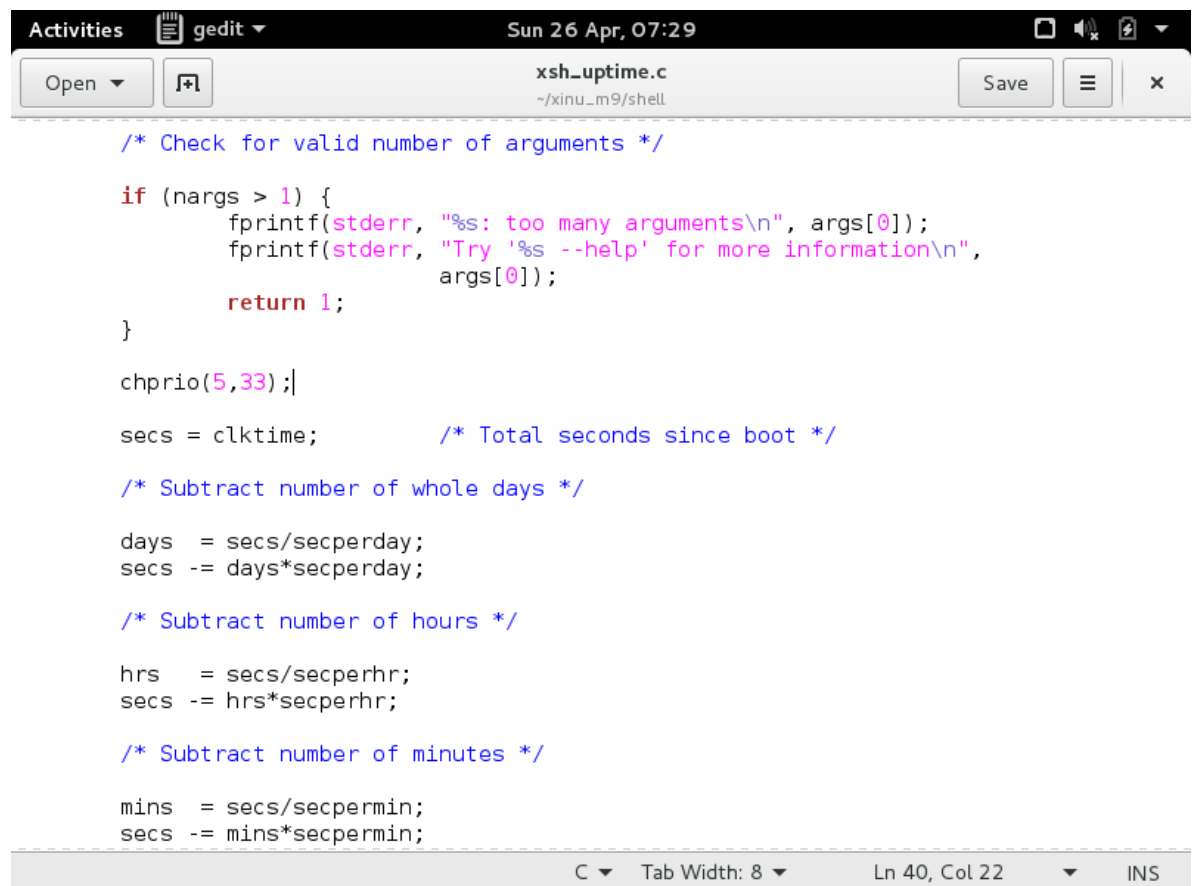
```

d. Save file

e. Compile dengan make clean lalu make

3. Tambahkan `chprio(5,33)` di `xsh_uptime.c`, compile, lalu jalankan `ps` → `uptime` → `ps` lagi untuk membuktikan prioritas `pid=5` berubah. Ulangi testing dengan nilai `-3` dan `id 3000` untuk membuktikan validasi error bekerja.

Jawaban:



```

Activities gedit Sun 26 Apr, 07:29
xsh_uptime.c
~/xinu_m9/shell

/* Check for valid number of arguments */

if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

chprio(5,33);

secs = clktime; /* Total seconds since boot */

/* Subtract number of whole days */

days = secs/secperday;
secs -= days*secperday;

/* Subtract number of hours */

hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */

mins = secs/secpermin;
secs -= mins*secpermin;

```

C Tab Width: 8 Ln 40, Col 22 INS

```
Activities Terminal Sun 26 Apr, 07:31
xinu@xinu-develop-end: ~/xinu_m9/compile
File Edit View Search Terminal Help

Obtained IP address 10.0.3.15 (0x0a00030f)

-----
XINU
-----

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process    recv   20   3 0x005E7FFC 0x005E7F44 65536
5 shell           recv   50   4 0x005D7FFC 0x005D79AC 8192
6 ps              curr   20   5 0x005F7FFC 0x005F7FC4 8192
xsh $

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

```
Activities Terminal Sun 26 Apr, 07:31
xinu@xinu-develop-end: ~/xinu_m9/compile
File Edit View Search Terminal Help
-- -- ---- - - ----
-----

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size
-----
 0 prnull         ready  0    0 0x005FDFFC 0x00FFFE4 8192
 1 ipout          wait   500  0 0x005FBFFC 0x005FBF30 8192
 2 netin          wait   500  0 0x005F9FFC 0x005F9ED0 8192
 4 Main process   recv    20  3 0x005E7FFC 0x005E7F44 65536
 5 shell         recv    50  4 0x005D7FFC 0x005D79AC 8192
 6 ps            curr    20  5 0x005F7FFC 0x005F7FC4 8192
xsh $ uptime

New Syscall is easy

Xinu has been up 3 second(s)

New Syscall is easy

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

```
Activities Terminal Sun 26 Apr, 07:31
xinu@xinu-develop-end: ~/xinu_m9/compile

File Edit View Search Terminal Help

-----
0 prnull      ready    0      0 0x005FDFFC 0x00FFFE4 8192
1 ipout       wait     500    0 0x005FBFFC 0x005FBF30 8192
2 netin       wait     500    0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv     20    3 0x005E7FFC 0x005E7F44 65536
5 shell       recv     50    4 0x005D7FFC 0x005D79AC 8192
6 ps          curr    20    5 0x005F7FFC 0x005F7FC4 8192
xsh $ uptime

New Syscall is easy

Xinu has been up 3 second(s)

New Syscall is easy

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready    0      0 0x005FDFFC 0x00FFFE4 8192
1 ipout       wait     20      0 0x005FBFFC 0x005FBF30 8192
2 netin       wait     500      0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv     20      3 0x005E7FFC 0x005E7F44 65536
5 shell       recv     33      4 0x005D7FFC 0x005D79AC 8192
8 ps          curr    20      5 0x005F7FFC 0x005F7FC4 8192
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

- Buka xsh_uptime.c di folder xinu_m9/shell/
- Tambahkan chprio(5,33) setelah blok if (nargs > 1)
- Save file
- Compile dengan make clean lalu make
- Jalankan sudo minicom
- Jalankan ps, catat pid=5 prio 50
- Jalankan uptime
- Jalankan ps lagi, pid=5 prio berubah jadi 33 ✓

Activities gedit Sun 26 Apr, 07:41

Open xsh_uptime.c Save

~/xinu_m9/shell

```
/* Check for valid number of arguments */

if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
              args[0]);
    return 1;
}

chprio(5,-3);

secs = clktime;          /* Total seconds since boot */

/* Subtract number of whole days */

days = secs/secperday;
secs -= days*secperday;

/* Subtract number of hours */

hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */

mins = secs/secpermin;
secs -= mins*secpermin;
```

C Tab Width: 8 Ln 40, Col 22 INS

```
Activities Terminal Sun 26 Apr, 07:43
xinu@xinu-develop-end: ~/xinu_m9/compile

File Edit View Search Terminal Help

Obtained IP address 10.0.3.15 (0x0a00030f)

-----
XINU
-----

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process    recv   20   3 0x005E7FFC 0x005E7F44 65536
5 shell           recv   50   4 0x005D7FFC 0x005D79AC 8192
6 ps              curr   20   5 0x005F7FFC 0x005F7FC4 8192
xsh $

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

```
Activities Terminal Sun 26 Apr, 07:43
xinu@xinu-develop-end: ~/xinu_m9/compile
File Edit View Search Terminal Help
-----
Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process    recv   20   3 0x005E7FFC 0x005E7F44 65536
5 shell           recv   50   4 0x005D7FFC 0x005D79AC 8192
6 ps              curr   20   5 0x005F7FFC 0x005F7FC4 8192

xsh $ uptime
Xinu has been up

New Syscall is easy

New Syscall is easy

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

Testing prioritas negatif chprio(5,-3):

- a. Ganti chprio(5,33) menjadi chprio(5,-3) di xsh_uptime.c
- b. Save file
- c. Compile dengan make clean lalu make
- d. Jalankan sudo minicom
- e. Jalankan ps, catat pid=5 prio 50
- f. Jalankan uptime
- g. Jalankan ps lagi, pid=5 prio tetap 50 (tidak berubah) ✓

Activities gedit Sun 26 Apr, 08:40

Open xsh_uptime.c Save

~/xinu_m9/shell

```
/* Check for valid number of arguments */

if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
              args[0]);
    return 1;
}

chprio(3000,3);

secs = clktime;          /* Total seconds since boot */

/* Subtract number of whole days */

days = secs/secperday;
secs -= days*secperday;

/* Subtract number of hours */

hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */

mins = secs/secpermin;
secs -= mins*secpermin;
```

C Tab Width: 8 Ln 40, Col 24 INS

```
Activities Terminal Sun 26 Apr, 08:34
xinu@xinu-develop-end: ~
File Edit View Search Terminal Help

Obtained IP address 10.0.3.15 (0x0a00030f)

-----
XINU
-----

Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv    20  3 0x005E7FFC 0x005E7F44 65536
5 shell       recv    50  4 0x005D7FFC 0x005D79AC 8192
6 ps          curr    20  5 0x005F7FFC 0x005F7FC4 8192
xsh $

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

```
Activities Terminal Sun 26 Apr, 08:34
xinu@xinu-develop-end: ~
File Edit View Search Terminal Help
-----
Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process    recv   20   3 0x005E7FFC 0x005E7F44 65536
5 shell           recv   50   4 0x005D7FFC 0x005D79AC 8192
6 ps              curr   20   5 0x005F7FFC 0x005F7FC4 8192
xsh $ uptime
Xinu has been up

New Syscall is easy

New Syscall is easy

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

```
Activities Terminal Sun 26 Apr, 08:35
xinu@xinu-develop-end: ~
File Edit View Search Terminal Help
-----
0 prnull      ready    0      0 0x005FDFFC 0x00FFFE4 8192
1 ipout       wait     500    0 0x005FBFFC 0x005FBF30 8192
2 netin       wait     500    0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv     20    3 0x005E7FFC 0x005E7F44 65536
5 shell       recv     50    4 0x005D7FFC 0x005D79AC 8192
6 ps          curr    20    5 0x005F7FFC 0x005F7FC4 8192
xsh $ uptime
Xinu has been up

New Syscall is easy

New Syscall is easy

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready    0      0 0x005FDFFC 0x00FFFE4 8192
1 ipout       wait     20      0 0x005FBFFC 0x005FBF30 8192
2 netin       wait     500      0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv     20      3 0x005E7FFC 0x005E7F44 65536
5 shell       recv     50      4 0x005D7FFC 0x005D79AC 8192
8 ps          curr    20      5 0x005F7FFC 0x005F7FC4 8192
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

Testing id tidak valid chprio(3000,3):

- a. Ganti chprio(5,-3) menjadi chprio(3000,3) di xsh_uptime.c
- b. Save file
- c. Compile dengan make clean lalu make
- d. Jalankan sudo minicom
- e. Jalankan ps, catat pid=5 prio 50
- f. Jalankan uptime
- g. Jalankan ps lagi, pid=5 prio tetap 50 (tidak berubah) ✓